S2.1 The ionic model

Formation of ions

Learning outcomes

By the end of this section you should be able to:

- Identify cations and anions from their neutral atom.
- Deduce the charge and symbol for ions.
- Deduce the electron configuration of a species based on its ionic symbol.

In section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) you learned that elements are the primary constituents of matter, which cannot be chemically broken down into simpler substances. Growing up, you may have learned that iron is important for your diet and you can likely name some foods that are rich in iron, for example, meat, beans and dark leafy greens. You may also have learned that iron is a metal used to create car bodies, bridges and skyscrapers (**Figure 1**). When eating iron rich foods, do you see or eat pieces of metal?

Credit: Anjelika Gretskaia, Getty Images (https://www.gettyimages.co.uk/detail/photo/fresh-spinach-leaves-on-white-background-royalty-free-image/1369248724)

Credit: tunart, Getty Images (https://www.gettyimages.co.uk/detail/photo/the-iron-bridges-of-chicago-downtown-royalty-free-image/1447903626)

Figure 1. Examples of foods and structures containing iron.

In section S1.2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/) you learned that all atoms of the same element have the same number of protons. You also learned that isotopes of the same element will have the same chemical properties, even though the number of neutrons differ. What subatomic particle could be responsible for the difference between the iron in foods we eat versus the iron we use in building materials? In section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) you learned about the electron configuration of atoms. Can there be a change to the number of electrons that might result in different properties for the same element?

Everything is made up of atoms, yet these atoms rarely exist as individual atoms. Atoms usually interact with other atoms to form chemical bonds. Chemical bonds create a more stable electron configuration for the atoms involved.

In previous sections, you have learned about elements and the nuclear and electronic components of the atom. Elements are the primary constituents of matter. Elements can be classified as metals or non-metals. **Figure 2** shows a periodic table classifying metals and non-metals. Note that there is also a region in between the two that are classified as metalloids. These are elements that show intermediate properties between metals and non-metals.

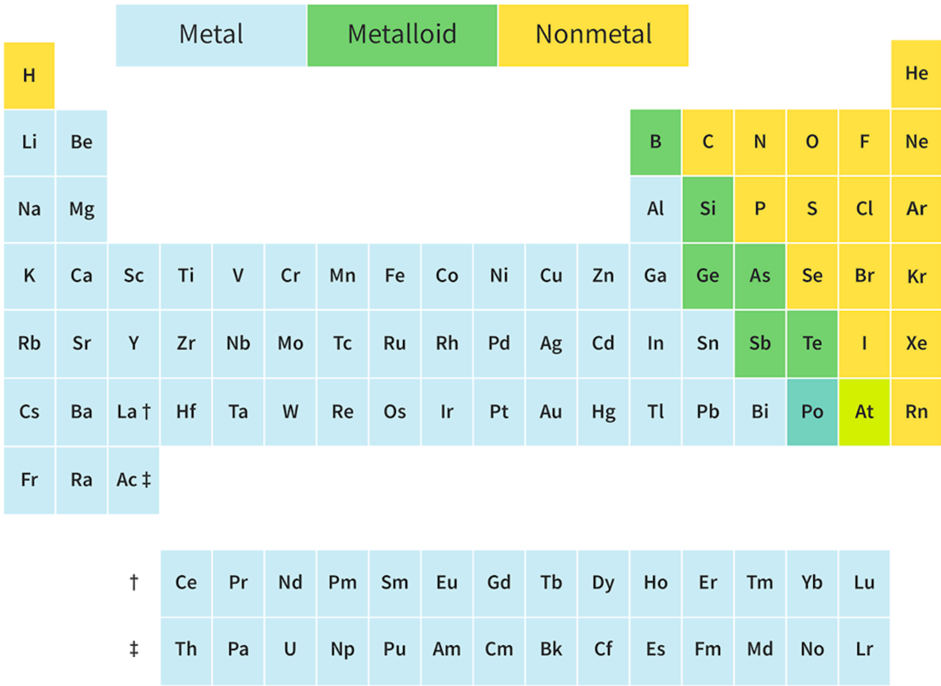

Figure 2. Classification of metals, non-metals and metalloids on the periodic table.

More information for figure 2

This diagram displays the periodic table of elements, categorizing them into metals, nonmetals, and metalloids. The metals are shaded in light blue, occupying the majority of the table, particularly on the left and center. Nonmetals are highlighted in yellow on the top right, including elements like hydrogen (H), carbon (C), nitrogen (N), and oxygen (O). Metalloids, bordered in green, sit between metals and nonmetals, featuring elements such as boron (B), silicon (Si), and arsenic (As). The diagram provides a visual representation of the classification of elements based on their properties.

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When looking at electron configurations in section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) what configurations were the most stable? Did you notice that these stable electron configurations occur when valence electrons occupy an energy level that is completely filled? What about atoms that have valence electrons occupying an energy level that is not completely filled? How might these atoms become more stable?

Metallic elements have low numbers of electrons in the valence shells. How do you think a metallic element might become more stable? Non-metals have a greater number of electrons in the valence shell. How do you think a non-metal might become more stable? A metal can lose electrons or a non-metal can gain electrons in order to become more stable forming an ion.

Exercises 1 and 2

Click a question to answer

1. ________________ has the electron configuration $1s^2 2s^2 2p^6$.

________________ has the unstable electron configuration $1s^2 2s^2 2p^6 3s^1$ or [Ne] $3s^1$.

2. True or false?

Neon has a stable electron configuration.

Check answers

Ions

An ion is an electrically charged species that is formed after an atom gains or loses an electron.

- When metal atoms lose electrons, they form positive ions called cations. A cation has fewer electrons than protons.
- When non-metal atoms gain electrons, they form negative ions called anions. An anion has more electrons than protons.

The number of protons and electrons in an atom are equal, and the atom is electrically neutral. The number of protons and electrons in an ion is not the same and the ion carries an overall charge.

Figure 3a shows a comparison of the electron structure for a neutral sodium atom and a sodium cation. **Figure 3b** shows a comparison of the electron structure for a neutral chlorine atom and a chloride anion (not to scale).

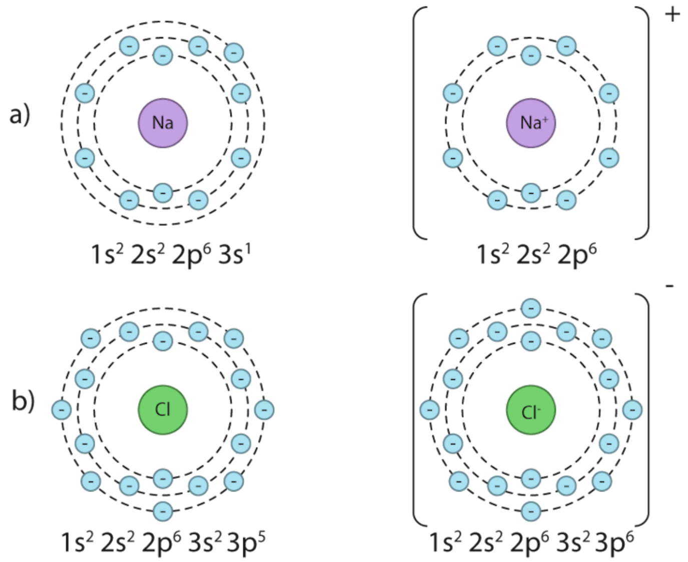

Figure 3. Comparison of the electron configuration of neutral atoms to ions: a) cations and b) anions.

More information for figure 3

The image shows two sets of electron configuration diagrams. In diagram a), on the left, a neutral sodium atom is depicted. It's represented by a central circle labeled 'Na' surrounded by circular shells showing electron distribution: $1s^2$, $2s^2$, $2p^6$, $3s^1$. On the right of this is a sodium cation (Na^+), which shows a central circle labeled 'Na^+', with a changed electron distribution excluding the outer electron: $1s^2$, $2s^2$, $2p^6$.

In diagram b), on the left, a neutral chlorine atom is represented by a central circle labeled 'Cl', with its electrons distributed in the shells as: $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^5$. On the right is a chloride anion (Cl^-) where the central circle is labeled 'Cl^-', with an added electron, completing the outer shell: $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$.

[Generated by AI]

Interactive 1 shows a visual representation of the formation of anions and cations.

Interactive 1. Visual representation of the formation of cations.

More information for interactive 1

An interactive animation model depicts the creation of a sodium cation. It illustrates the atomic structure of a sodium atom, displaying 11 electrons surrounding the nucleus and 11 neutrons in the nucleus. A seek bar with a play/pause button is at the bottom. Selecting the play option shows an electron leaving the sodium atom, transforming it into a sodium ion (Na^{+}). A toggle button for repeat and skip forward is at the right of the slider.

A panel below the title-cation creation on the left provides view options, including the entire model, electrons, and protons. Selecting electrons or protons reveals their arrangement within the structure. On the right side, the contrast or dark mode and font settings are available. At the bottom, options for introduction, share, screenshot, augmented reality, and language settings are available.

The introduction reads as follows:

Cations are positively charged particles, which can be either atoms or molecules. They have one or more electrons fewer than their neutral form. The model depicts the formation of sodium cations, according to the reaction: $Na \rightarrow Na^{+} + e^{-}$

Ionic compounds are composed of both cations (positively charged ions) and anions. These ions are held together by strong electrostatic forces of attraction.

The model illustrates the transformation of a sodium atom into a sodium cation and depicts the electron arrangement before and after ion formation.

Interactive 2. Visual representation of the formation of anions.

More information for interactive 2

An interactive animation model depicts the creation of a chloride anion. It illustrates the atomic structure of a chlorine atom, displaying 17 electrons surrounding the nucleus and 17 neutrons in the nucleus. A seek bar with a play/pause button is at the bottom.
Selecting the play option shows an electron approaching from the top left, entering the chlorine atom, transforming it into a chlorine ion (Cl^{-}). A toggle button for repeat and skip forward is at the right of the slider.
A panel below the title- anion creation on the left provides view options, including the entire model, electrons, and protons. Selecting electrons or protons reveals their arrangement within the structure. On the right side, the contrast or dark mode and font settings are available.
At the bottom, options for introduction, share, screenshot, augmented reality, and language settings are available.
The introduction reads as follows:
Anions are negatively charged particles, which can be either atoms or molecules. They have one or more extra electrons than their neutral form. The model depicts the formation of chlorine anions, according to the reaction: $Cl + e^{-} \rightarrow cl^{-}$
Ionic compounds are composed of both cations (positively charged ions) and anions. These ions are held together by strong electrostatic forces of attraction.
The model illustrates the transformation of a chlorine atom into a chloride anion and depicts the electron arrangement before and after ion formation.

Making connections

When an ion is formed, its radius changes from its atom. The ionic radius of an anion is always larger than its neutral atomic radius and the ionic radius of a cation is always smaller than its neutral atomic radius. This is covered in section S3.1.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/).

The octet rule refers to the tendency of an atom to have a valence shell with a total of 8 electrons. You will find more detail about this rule in section S2.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/). This rule is helpful in deducing how many electrons an atom will lose or gain in order to become electronically stable. **Table 1** shows the deduction of ionic charge for Period 3 elements. Note that this pattern is applicable to the entire group.

Table 1. Deducing the ionic charge for Period 3 elem

Group number	1	2	13	14	15	16
Period 3 elements	Na	Mg	Al	Si	P	S
Electron configuration	[Ne] $3s^1$	[Ne] $3s^2$	[Ne] $3s^2 3p^1$	[Ne]$3s^2 3p^2$	[Ne] $3s^2 3p^3$	[Ne] $3s^2 3p^4$
Number of electrons in valence shell	1	2	3	4	5	6
Electron gain/ loss	Lose 1	Lose 2	Lose 3	Unlikely to form ions	Gain 3	Gain 2
Ion formed	Na^+	Mg^{2+}	Al^{3+}	N/A	P^{3-}	S^{2-}
Electron configuration of ion	[He] $2s^2 2p^6$	[He] $2s^2 2p^6$	[He] $2s^2 2p^6$		[Ne] $3s^2 3p^6$	[Ne] $3s^2 3p^6$

Since sodium has one valence electron, it is energetically more favourable to lose one electron than to gain seven to achieve a full valence shell. After losing one electron, it will have more protons in the nucleus than electrons surrounding it.

This will give it an overall charge of 1+. Notationally, the symbol for an ion is represented by the element symbol first, followed by a superscript to show the charge. In the superscript, the number is always written first, followed by the charge. This means that when the neutral element Na loses an electron, it would be represented by Na^{1+}, or more commonly Na^{+}.

The box below has some important things to consider for proper notation of ions.

Study skills

Notation for ions

1. (a) Ions are expressed using elemental symbols stated in the guide and section 7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) of the DP chemistry data booklet. The charge of an ion can be deduced by considering the ratio of protons to electrons.
(b) When ions form, it is electrons that are either lost or gained. The number of protons is **always** the same for a particular atom.
2. The number is **always** written first, followed by charge for the symbolic form of an ion, e.g. Mg^{2+}. When discussing the term oxidation state the charge is written first, e.g. The Mg^{2+} ion has a +2 oxidation state.
3. For an ion with a charge of +1 or —1, the 1 does not need to be shown in the element symbol. For example, lithium forms an ion with a charge of 1+. The symbol for this ion would be Li^{+}.

As sulfur has six valence electrons, it is energetically more favourable to gain two electrons than to lose six. After gaining two electrons, it will have more electrons surrounding it than protons in the nucleus, giving it an overall charge of 2–. This can be represented as S^{2-}. This means that you can predict the magnitude and sign of the charge of an ion that is formed from an atom of an element in a specific group.

Nature of Science

Aspect: Patterns and trends

The placement of atoms within groups in the periodic table allows students and scientists to easily deduce how many valence electrons each element has, and therefore, the ion that they are likely to form (or even if they are likely to form an ion). Atoms that form more than one ion with different charges are also located within the same region of the periodic table.

You might have noticed that in **Table 1** silicon and argon were not assigned a charge. This is because silicon has four valence electrons, giving it a low tendency to form ions. Argon already has a full valence shell with a stable electron configuration and also has a low tendency to form ions.

Transition elements

Transition elements are those located in groups 3–12 on the periodic table. In section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) you learned how to deduce electron configuration of elements and ions up to $Z = 36$. What sublevel are valence electrons occupying for $Z = 21$ to $Z = 30$? Recall how the 3d sublevel is very close in energy to the 4s and 4p sublevels (illustrated in **Figure 4**). This allows electrons from both the 4s and 3d to act as valence electrons for elements $Z = 21$ to $Z = 30$.

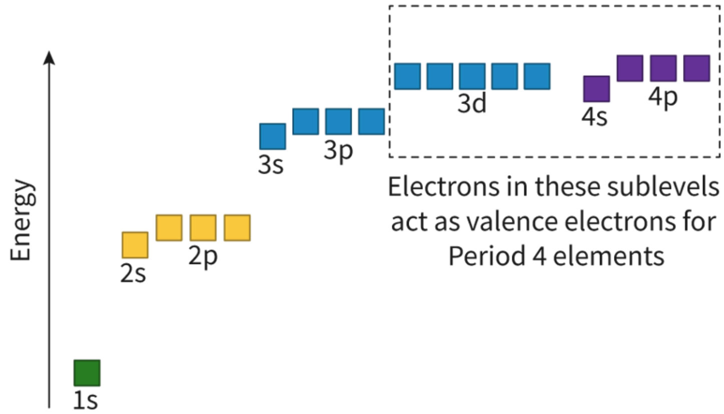

Figure 4. Relative energies of sublevels of the atom.

More information for figure 4

The image is a diagram illustrating the relative energies of atomic sublevels ranging from 1s to 4p. The diagram consists of horizontal blocks at different energy levels on a vertical axis labeled 'Energy.' Starting from the bottom, there is a single green block labeled '1s,' followed by two yellow blocks labeled '2s' and '2p.' Above these are three blue blocks labeled '3s,' '3p,' and '3d.' At the top, there are two purple blocks labeled '4s' and '4p.' The 3d, 4s, and 4p blocks are enclosed in a dashed rectangle, with a note indicating that electrons in these sublevels act as valence electrons for Period 4 elements. The blocks and their alignment are used to visually represent the close energy relationship shared among the sublevels 4s, 3d, and 4p.

[Generated by AI]

When transition elements in Period 4 lose electrons to form a cation, electrons are first lost from the 4s sublevel followed by the 3d sublevel. These two different sublevels where electrons are lost result in some variability in the charge of the ion formed for transition elements. Let's consider two possible titanium ions below:

- Ti^{2+} two electrons lost from the 4s sublevel.

- Ti^{4+} two electrons lost from the 4s sublevel and two electrons lost from the 3d sublevel.

The interactive below shows the comparison of the electron configurations between neutral titanium and the two ions discussed above.

Select, using the buttons, the different types of titanium. Look at each electron configuration and orbital diagram in **Interactive 2**.

Interactive 2. Electron configurations and orbital diagrams for Ti, Ti^{2+} and Ti^{4+}.

More information for interactive 2

An interactive tool shows orbital diagrams for Ti, Ti2+, and Ti4+, along with their electronic configurations. Three buttons at the bottom, labeled Ti, Ti2+, and Ti4+, each have a corresponding checkbox. Selecting a titanium species displays the appropriate orbital diagram and electron configuration.
A vertical arrow on the left represents energy. The energy level increases in the following order: 1s, 2s, 2p, 3s, 3p, 4s, 3d, and 4p. In the orbital diagram, the s orbital has a single box, the p orbital has three boxes, and the d orbital has five boxes. Electrons are represented by arrows inside these boxes. Each box represents an orbital. The 3d, 4s, and 4p orbitals are represented as valence electrons.
Titanium (Ti) has 22 electrons, with an electron configuration of 1s2 2s2 2p6 3s2 3p6 4s2 3d2. The 1s, 2s, 3s, and 4s orbitals have two electrons in opposite spins, 2p and 3p orbitals have six electrons, with each box having two electrons in opposite directions, and the 3d orbital has two electrons in the first two boxes. The 4p orbital is empty.
By selecting the Ti2+, its orbital diagram and electronic configuration are displayed. Titanium (Ti 2+) has 20 electrons, with an electron configuration of 1s2 2s2 2p6 3s2 3p6 3d2. Two electrons are removed from the 4s orbital of the neutral titanium atom.

By selecting the Ti 4+, its orbital diagram and electronic configuration are displayed. Titanium (Ti 4+) has 18 electrons, with an electron configuration of 1s2 2s2 2p6 3s2 3p6. Two electrons are removed from the 4s orbital, and two electrons are removed from the 3d orbital of the neutral titanium atom.
This interactive tool helps users understand the electron arrangement in neutral titanium (Ti) and its ions (Ti^{2+} and Ti^{4+}).

Table 2 shows the electron configurations for some of the common ions formed from transition elements in Period 4.

Table 2. Electron configuration and common ionic charges for Period 4 transition eleme

Element	Electron configuration of neutral element	Common ions	Common i
Cr	$[Ar]4s^1 3d^5$	Cr^{2+} Cr^{3+}	$[Ar]3d^4$ $[Ar]3d^3$
Fe	$[Ar]4s^2 3d^6$	Fe^{2+} Fe^{3+}	$[Ar]3d^6$ $[Ar]3d^5$
Cu	$[Ar]4s^1 3d^{10}$	Cu^{+} Cu^{2+}	$[Ar]3d^{1C}$ $[Ar]3d^9$

Use the following activity to test your understanding of electron configuration of ions.

Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:**
  - Thinking skills
    - Critically evaluating information to reach a conclusion
    - Metacognitive awareness of how understanding is applied
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity to start, followed by work in pairs

Download and complete the following worksheet.

Worksheet
(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry

Once you have completed the worksheet, compare your chart with a peer. Discuss with your peer which rows you found straightforward to complete and which rows you found difficult to complete. Why do you think some rows were easier than others?